BTS-VI(SS)-01.17-0096 Reg.No. | | | | | | | | |

B

B.Tech. Degree VI Semester Special Supplementary Examination January 2017

ME 1605 CAD/CAM

(2012 scheme)

Time : 3 Hours Maximum Marks : 100

PART A

(Answer *ALL* questions)

$(8 \times 5 = 40)$

I. (a) Discuss different geometrical modeling techniques.

(b) Write notes on Detroit Type of Automation.

(c) Differentiate between open loop and closed loop control systems.

(d) Differentiate between miscellaneous and preparatory functions used in manual part programming.

(e) Distinguish between static and dynamic errors.

(f) Illustrate the difference between tool magazine and tool turret.

(g) Define work envelope of an industrial robot with a proper sketch.

(h) Write short notes on FMS and CIM.

PART B

$(4 \times 15 = 60)$

II. Discuss different transfer mechanisms used to move parts between stations in AFL with suitable sketches. (15)

OR

III. The table below defines the precedence relationship and element times for a new model. Construct the precedence diagram for this job. The ideal cycle time = 1.1 min repositioning time = 0.1 min and uptime proportion is assumed to be 1.0. Use the Ranked positional weights rule to assign work element to stations. (15)

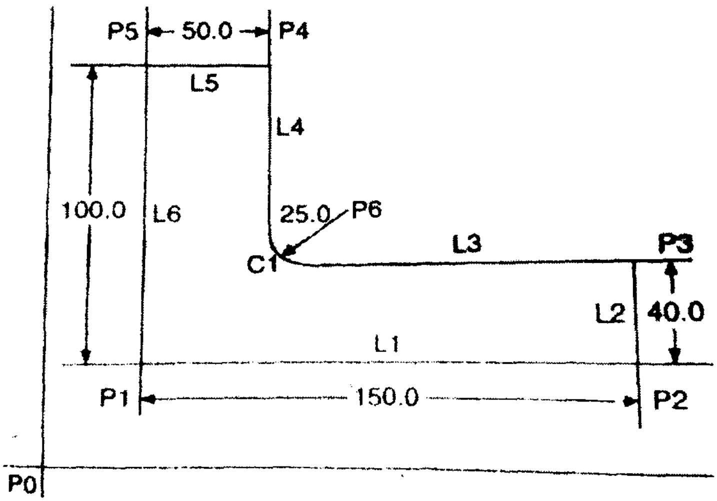
Work element	Te (min)	Immediate predecessors
1	0.5	-
2	0.3	1
3	0.8	1
4	0.2	2
5	0.1	2
6	0.6	3
7	0.4	4,5
8	0.5	3,5
9	0.3	7,8
10	0.6	6,9

IV. (a) What are the various classifications of CNC machines? Explain with sketches. (10)

(b) Explain absolute and incremental dimensioning in CNC machines with the help of suitable figures. (5)

OR

V. Write a detailed APT geometric statement to derive the outline of the cam shown in the figure with a cutter diameter of 6.5 mm, cutting speed of 900 rpm and feed rate 7.5 mm/min. (15)



VI. What are the different areas where design changes are required in CNC machines? Explain in detail with reasons for each area. (15)

OR

VII. (a) Discuss about various testing methods of CNC machine tool performance. (10)

(b) Differentiate between a turning centre and machining centre. (5)

VIII. (a) Describe the common types of robot configurations with neat sketches. (10)

(b) Differentiate between PTP and CP system in robots. (5)

OR

IX. (a) What are AI and expert systems for manufacturing automation? (8)

(b) Explain the application of industrial robots in manufacturing industry. (7)

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